

Day 1 Notes : Introduction to DBMS

DBMS Notes

1. What is DBMS?

DBMS stands for **Database Management System**.

It is software that is used to **store, organize, manage, retrieve, and update data efficiently**.

A DBMS acts as an interface between the user/application and the database.

For example, a college database may contain information about:

- Students
- Teachers
- Courses
- Marks
- Attendance
- Fees

Instead of storing all this information in separate files, a DBMS stores and manages it in an organized manner.

Examples of DBMS are:

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server
- MongoDB

Interview Answer

A DBMS is software that allows users to store, retrieve, update, and manage data efficiently while providing features like security, consistency, concurrency, and backup.

2. Why is DBMS Needed?

Before DBMS, data was mainly stored using traditional file systems.

As the amount of data increased, file systems created several problems such as:

- Duplicate data
- Data inconsistency
- Difficult data searching
- Weak security
- Difficult backup and recovery
- Problems when multiple users access data simultaneously

DBMS was introduced to solve these problems.

Main Reasons for Using DBMS

1. Easy Data Storage

A DBMS allows large amounts of data to be stored in an organized manner.

2. Easy Data Retrieval

Users can easily search and retrieve required information from the database.

3. Reduced Data Redundancy

The same information does not need to be stored repeatedly.

4. Better Data Consistency

When data is updated, DBMS ensures that correct and consistent information is maintained.

5. Security

DBMS allows different levels of access for different users.

For example:

- Administrator can read, update, and delete data.
- Employee may only read and update selected information.
- Customer may only access their own information.

6. Concurrent Access

Multiple users can access the same database at the same time.

For example, thousands of users can simultaneously use an online shopping or banking application.

7. Backup and Recovery

DBMS provides mechanisms to recover data if the system crashes or data is accidentally lost.

3. DBMS vs File System

File System

In a traditional file system, data is stored directly inside files.

For example:

Student.txt

Employee.txt

Orders.txt

Managing a large amount of data through separate files becomes difficult.

DBMS

In DBMS, data is stored in a structured database and is managed using database software.

Major Differences

Data Redundancy

File System: High possibility of duplicate data.

DBMS: Redundancy can be reduced.

Data Consistency

File System: Different files may contain different values for the same information.

DBMS: Better data consistency is maintained.

Security

File System: Limited security.

DBMS: Provides user authentication and access control.

Searching

File System: Searching large amounts of data is difficult.

DBMS: Data can be searched easily.

Relationships

File System: Maintaining relationships between different files is difficult.

DBMS: Relationships between tables can easily be created.

Concurrent Access

File System: Difficult to handle multiple users.

DBMS: Supports multiple users simultaneously.

Backup and Recovery

File System: Backup usually has to be handled manually.

DBMS: Provides backup and recovery facilities.

Transactions

File System: Does not properly support transactions.

DBMS: Provides transaction management.

Interview Answer

A file system stores information directly in files, whereas a DBMS stores and manages data in a structured way. DBMS provides better security, consistency, concurrency, relationships, transactions, and backup compared to a traditional file system.

4. Advantages of DBMS

1. Reduced Data Redundancy

Data redundancy means storing the same information multiple times.

A DBMS reduces unnecessary duplication of data.

2. Improved Data Consistency

If the same information is stored in multiple places, different copies may contain different values.

DBMS helps maintain consistent information.

3. Better Security

Only authorized users can access the database.

Permissions can be provided according to the user's role.

4. Data Integrity

Data integrity means maintaining correct and valid data.

DBMS can apply rules such as:

- Student ID must be unique.
 - Email cannot be duplicated.
 - Employee ID cannot be empty.
 - A department must exist before assigning an employee to it.
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5. Multiple User Access

Multiple users can access the database simultaneously without corrupting data.

6. Backup and Recovery

DBMS can recover information after:

- System crash
 - Power failure
 - Hardware failure
 - Accidental deletion
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7. Better Data Sharing

Different departments of an organization can share the same database.

For example:

- HR
- Finance
- Sales
- Management

can use the same employee database.

8. Transaction Management

DBMS ensures that important operations are completed correctly.

For example, during a bank transfer:

₹1,000 must be deducted from Account A and ₹1,000 must be added to Account B.

Both operations should either happen successfully or neither should happen.

5. Types of Databases

There are different types of databases depending on how and where data is stored.

Important types are:

1. Relational Database

2. Non-Relational Database
 3. Centralized Database
 4. Distributed Database
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6. Relational Database

A relational database stores data in the form of **tables**.

Each table contains:

- Rows
- Columns

For example, a Student table may contain:

Student ID
Name
Course
Marks
Email

Each row represents one student and each column represents an attribute of the student.

Different tables can also be connected using keys.

Examples of relational databases:

- MySQL
- PostgreSQL
- Oracle
- SQL Server

Main Characteristics

- Data is stored in tables.
 - Tables can have relationships.
 - Generally uses SQL.
 - Provides strong data consistency.
 - Suitable for structured data.
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7. Non-Relational Database

A non-relational database is also commonly known as a **NoSQL database**.

It does not necessarily store information in traditional rows and columns.

It is useful when:

- Data structure changes frequently.
- Very large amounts of data have to be handled.
- Flexible data storage is required.

Examples include:

- MongoDB
- Redis
- Cassandra
- Neo4j

Types of NoSQL Databases

Document Database

Stores data in documents.

Example: MongoDB.

Key-Value Database

Data is stored as a key and its associated value.

Example: Redis.

Graph Database

Stores data in the form of nodes and relationships.

Example: Neo4j.

Column-Based Database

Stores information using column families.

Example: Cassandra.

8. Relational vs Non-Relational Database

Relational Database

- Stores data in tables.
- Has rows and columns.
- Usually follows a fixed schema.
- Uses SQL.
- Good for highly structured data.
- Examples: MySQL, PostgreSQL and Oracle.

Non-Relational Database

- Can store data as documents, graphs, key-value pairs, etc.
 - Usually has a flexible schema.
 - Suitable for large and rapidly changing data.
 - Examples: MongoDB, Redis and Cassandra.
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9. Centralized Database

A centralized database stores the complete database at **one central location**.

All users connect to the same central database server.

For example, an organization may maintain all employee information on one database server located at its head office.

Advantages

- Easy to manage.
- Easy to maintain.
- Easier backup.
- Better control over security.
- Data remains consistent.

Disadvantages

- Central server becomes a single point of failure.
- If the central server fails, the complete system may become unavailable.
- Heavy traffic may reduce performance.

10. Distributed Database

A distributed database stores data across **multiple physical locations or servers**.

These servers may be located in different:

- Cities
- Countries
- Data centers

Even though data is distributed, users may see it as one database.

For example, a company may have database servers in:

- Delhi
- Mumbai
- Bengaluru

Advantages

- Better availability.
- Better fault tolerance.
- Faster access for geographically distributed users.
- Failure of one server may not stop the complete system.

Disadvantages

- More difficult to manage.
- Data synchronization is complex.
- Security is more difficult.
- Maintaining consistency between servers can be challenging.

11. DBMS Architecture

DBMS architecture describes how users, applications, and databases interact with each other.

The major architectures are:

1. One-Tier Architecture

2. Two-Tier Architecture
 3. Three-Tier Architecture
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12. One-Tier Architecture

In one-tier architecture, the user directly interacts with the database.

There is no separate application server between the user and database.

It is mostly used for:

- Local application development
- Testing
- Database development

It is simple but not suitable for large applications.

13. Two-Tier Architecture

Two-tier architecture contains two main layers:

1. Client
2. Database Server

The client application communicates directly with the database server.

For example:

A desktop application may directly connect with a MySQL database.

Advantages

- Simple architecture.
- Faster for small applications.
- Easy to implement.

Disadvantages

- Less secure.

- Difficult to scale.
 - Client directly accesses the database.
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14. Three-Tier Architecture

Three-tier architecture contains three layers:

1. Presentation Layer
2. Application Layer
3. Database Layer

Presentation Layer

This is the interface used by the user.

For example:

- Website
- Mobile application
- Desktop application

Application Layer

This contains the business logic of the application.

It processes user requests before communicating with the database.

Database Layer

This is responsible for storing and managing data.

Advantages

- Better security.
- Better scalability.
- Easy maintenance.
- Business logic is separated from the user interface.
- Commonly used in modern web applications.

Interview Answer

In three-tier architecture, the client communicates with an application server, and the application server communicates with the database. This improves security, scalability, and maintainability.

15. Three-Level Schema Architecture

Another important DBMS architecture consists of three levels:

1. External Level
 2. Conceptual Level
 3. Internal Level
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External Level

The external level describes how different users view the database.

Different users may see different parts of the same database.

For example:

HR may see:

- Employee name
- Salary
- Department

Normal employees may only see:

- Employee name
 - Department
 - Designation
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Conceptual Level

The conceptual level describes the complete logical structure of the database.

It defines:

- Tables
- Attributes
- Relationships
- Constraints

It does not describe how data is physically stored.

Internal Level

The internal level describes how data is physically stored inside the system.

It includes:

- Files
 - Memory blocks
 - Indexes
 - Storage techniques
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16. Data Models in DBMS

A data model defines how information is organized and how relationships between data are represented.

Important data models are:

1. Hierarchical Model
 2. Network Model
 3. Relational Model
 4. Entity-Relationship Model
 5. Object-Oriented Model
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17. Hierarchical Data Model

In the hierarchical model, data is organized in a **tree structure**.

It follows a parent-child relationship.

One parent can have multiple children, but generally one child has only one parent.

Example:

Company
Department
Employee

The company contains departments, and departments contain employees.

18. Network Data Model

The network model is similar to the hierarchical model, but one child can have multiple parents.

It allows more complex relationships.

For example:

A student may belong to multiple courses and multiple clubs.

19. Relational Data Model

The relational model stores information in the form of tables.

Tables consist of:

- Rows
- Columns

Relationships between tables are created using keys.

It is the most commonly used database model.

20. Entity-Relationship Model

The Entity-Relationship model represents information using:

- Entities
- Attributes
- Relationships

Entity

An entity represents a real-world object.

Examples:

- Student
- Teacher
- Employee
- Product

Attribute

An attribute describes an entity.

For a Student entity, attributes may be:

- Student ID
- Name
- Age
- Course

Relationship

A relationship describes how entities are connected.

For example:

Student **enrolls in** Course.

Teacher **teaches** Course.

21. Schema

A **schema** is the overall structure or blueprint of a database.

It defines:

- Tables
- Column names
- Data types
- Keys
- Relationships
- Constraints

For example, a Student table may contain:

- Student ID
- Name
- Email
- Marks

This structure is part of the database schema.

Schema normally does not change frequently.

Interview Answer

Schema represents the logical structure or design of the database.

22. Instance

An **instance** represents the actual data present inside a database at a particular point in time.

For example, the Student table structure may remain the same, but the number of students and their marks may change every day.

Therefore:

Schema = Structure of database

Instance = Actual data present at a particular time

23. Metadata

Metadata means **data about data**.

It provides information about the structure and properties of data.

For example, consider a column called Student ID.

Its metadata may include:

- Column name: Student ID
- Data type: Integer
- Constraint: Primary Key
- Null allowed: No

The actual Student ID such as 101 is data.

The information describing Student ID is metadata.

Interview Answer

Metadata is information that describes other data, such as column names, data types, constraints, and table structure.

24. Keys in DBMS

Keys are attributes or combinations of attributes used to:

- Identify records uniquely.
- Establish relationships between tables.

Important keys are:

1. Primary Key
 2. Foreign Key
 3. Candidate Key
 4. Super Key
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25. Primary Key

A primary key is an attribute or set of attributes that **uniquely identifies every record in a table**.

For example:

In a Student table:

Student ID
Name
Email
Marks

Student ID can be selected as the primary key.

Properties of Primary Key

- It must contain unique values.
- It cannot contain NULL values.
- A table has only one primary key.
- It may consist of one or multiple columns.

Interview Answer

A primary key uniquely identifies each row in a table and cannot contain duplicate or NULL values.

26. Foreign Key

A foreign key is an attribute in one table that refers to the primary key of another table.

It is mainly used to **create relationships between tables**.

For example:

Suppose we have two tables:

Department Table:

- Department ID
- Department Name

Employee Table:

- Employee ID
- Employee Name
- Department ID

Department ID is the primary key in the Department table.

Department ID present in the Employee table becomes a foreign key.

Important Points

- Foreign keys can contain duplicate values.
- Foreign keys may contain NULL values depending on the database design.
- One table can contain multiple foreign keys.
- Foreign keys help maintain referential integrity.

Interview Answer

A foreign key is an attribute in one table that references the primary key of another table and is used to establish relationships between them.

27. Candidate Key

A candidate key is a **minimal attribute or set of attributes that can uniquely identify every row**.

Suppose a Student table contains:

- Student ID
- Email
- Aadhaar Number
- Name

If Student ID, Email and Aadhaar Number are all unique, all three can be candidate keys.

From these candidate keys, one is selected as the primary key.

Interview Answer

A candidate key is a minimal set of attributes capable of uniquely identifying every row in a table.

28. Super Key

A super key is any attribute or combination of attributes that can uniquely identify a row.

Suppose Student ID itself uniquely identifies each student.

Then all of these can be super keys:

- Student ID
- Student ID + Name
- Student ID + Email
- Student ID + Name + Email

The additional attributes may not be necessary, but the combinations can still uniquely identify a row.

Super Key vs Candidate Key

A candidate key is a **minimal super key**.

For example:

Student ID + Name may be a super key.

But if Student ID alone can uniquely identify the student, then Name is unnecessary.

Therefore:

Student ID = Candidate Key

Student ID + Name = Super Key but not Candidate Key

29. Relation Between Super Key, Candidate Key and Primary Key

The easiest way to remember this is:

Super Key → Candidate Key → Primary Key

A super key can uniquely identify a row.

A candidate key is a minimal super key.

One candidate key is selected as the primary key.

For example:

Suppose Student ID and Email both uniquely identify students.

Candidate Keys:

- Student ID
- Email

If Student ID is selected for the table, then:

Primary Key = Student ID

30. Database Users and Roles

Different types of users interact with databases.

Important database users are:

1. Database Administrator
 2. Database Designer
 3. Application Developer
 4. End User
 5. Data Analyst
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31. Database Administrator – DBA

A Database Administrator is responsible for managing and maintaining the complete database system.

Responsibilities of DBA

- Installing database software.
- Creating users.
- Providing permissions.
- Managing security.
- Creating backups.
- Recovering lost data.
- Monitoring database performance.

- Optimizing database performance.
- Maintaining database availability.

Interview Answer

A DBA is responsible for the security, availability, backup, recovery, performance, and overall management of the database.

32. Database Designer

A database designer decides how the database should be structured.

They decide:

- What tables are required.
- What columns should be present.
- What keys should be used.
- What relationships should exist.
- What constraints should be applied.

For example, in an e-commerce system, the designer may create tables for:

- Customer
 - Product
 - Order
 - Payment
 - Delivery
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33. Application Developer

Application developers build software that communicates with databases.

They are responsible for:

- Taking input from users.
- Sending requests to the backend.
- Reading information from databases.
- Updating database records.

- Displaying information to users.

For example, when a user places an order on an e-commerce website, the application developer creates the logic that saves that order in the database.

34. End Users

End users are people who use applications that indirectly interact with databases.

Examples:

- Customer using Amazon
- Student using college portal
- Employee using HR portal
- Customer using banking application

They normally do not directly interact with the database.

35. Data Analyst

A data analyst retrieves and analyzes data to provide useful business insights.

Their responsibilities include:

- Extracting data.
- Cleaning data.
- Analyzing trends.
- Calculating KPIs.
- Creating reports.
- Creating dashboards.
- Helping businesses make decisions.

They frequently work with:

- SQL
- Excel
- Power BI
- Tableau
- Python

Important Interview Revision

What is DBMS?

DBMS is software used to store, organize, retrieve, update and manage data efficiently.

Why do we need DBMS?

DBMS helps reduce redundancy, maintain consistency, provide security, support multiple users, and provide backup and recovery.

DBMS vs File System?

A file system stores information directly in files, whereas DBMS provides structured storage along with security, relationships, transactions, concurrency, backup and recovery.

What is Schema?

Schema is the logical structure or blueprint of a database.

What is Instance?

Instance is the actual data present in a database at a particular point in time.

What is Metadata?

Metadata means data about data. It describes information such as table names, column names, data types and constraints.

What is a Primary Key?

A primary key uniquely identifies every record in a table and cannot contain duplicate or NULL values.

What is a Foreign Key?

A foreign key references the primary key of another table and is used to establish a relationship between tables.

What is a Candidate Key?

A candidate key is a minimal set of attributes that can uniquely identify every row.

What is a Super Key?

A super key is any attribute or combination of attributes capable of uniquely identifying a row.

Candidate Key vs Super Key?

A candidate key is a minimal super key. A super key may contain additional unnecessary attributes.

What is Three-Tier Architecture?

Three-tier architecture consists of the presentation layer, application layer and database layer. It provides better security, scalability and maintainability.